

Tokenization & Quantum Risk: A Briefing

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Introduction

Tokenization representing financial assets, deposits, and payment credentials as programmable digital tokens is rapidly becoming core infrastructure across banking, payments, and capital markets.

At the same time, quantum computing is advancing toward a point where it could break public-key cryptography (RSA and elliptic-curve) that secures nearly all of this infrastructure today. The overlap of these two trends creates a distinct risk category: cryptographic assets and settlement systems being built now, on encryption that may not hold for their full operational life.

This brief pulls together the current state of that risk and the response to it, drawing on central bank research, industry readiness surveys, and live institutional pilots. It is organized in five parts: the official-sector regulatory response, industry-wide readiness data, and implementation progress across asset tokenization, payment tokenization, and the underlying blockchain/protocol layer.

The picture that emerges is one of a widening gap; quantum-safe migration is technically demonstrated in pilot form at leading institutions, but certification standards, hardware validation, and broad institutional readiness are lagging behind.

Executive Summary

Quantum computing is emerging as a structural risk to the cryptographic foundations of tokenized finance, from wholesale central bank settlement rails to retail payment cards and public blockchains. Central banks and standard-setters have moved from awareness to active experimentation and roadmap-building over the past 18 months [1] [2][3], while industry surveys show a persistent readiness gap: the majority of financial institutions still lack post-quantum cybersecurity plans despite widespread recognition that action is needed within 2–5 years [8][9].

Meanwhile, individual institutions such as HSBC, Mastercard, SWIFT, and a cluster of specialized quantum-security vendors are running live pilots that demonstrate migration is technically feasible but operationally complex [11][12][14][15].

1. Central Bank & Regulatory Landscape

The Bank for International Settlements has taken the lead in coordinating an official sector response. Its Project Leap initiative, run through the BIS Innovation Hub's Eurosystem Center

with the Bank of France, Bundesbank, and Banca d'Italia, has completed two phases of technical experimentation, successfully executing liquidity transfers using quantum-resistant digital signatures on TARGET2-like payment infrastructure [1][2].

The accompanying BIS Papers No. 158, "Quantum-readiness for the financial system: a roadmap," provides the first comprehensive migration framework for the sector and explicitly flags that stablecoins, tokenized deposits, and real-world assets (RWAs) integrating with traditional rails will need to align with post-quantum security standards [3].

This work has since been elevated to a formal multilateral track. The G7 Quantum Technology Working Group (QTWG), co-led by the Bank of France and Bank of Canada with the Federal Reserve, ECB, Bank of England, and Bank of Japan as members, published its first report in 2026, marking the beginning of systematic incorporation of quantum risk into financial stability assessment frameworks [4].

In parallel, the G7 Cyber Expert Group adopted a post-quantum cryptography transition roadmap in January 2026, targeting payment systems, central bank infrastructure, and systemically important service providers [5].

The IMF has focused more on tokenization's structural risk profile than quantum specifically, warning that atomic settlement and automation could cause stress events to unfold faster than regulators can respond, and calling for clearer legal frameworks and stronger cross-border coordination [6][7].

2. Industry Readiness Surveys

The clearest quantitative signal on institutional readiness comes from Moody's and Corinium Global Intelligence, which surveyed 200 data, analytics, and innovation leaders across 17 European and North American financial services and banking firms. The results are stark: 86% of organizations report they are not ready for post-quantum cybersecurity, while 84% expect the need to become acute within two to five years [8] [9].

This gap between recognized urgency and operational readiness recurs across the sector and is echoed by the Global Risk Institute's annual Quantum Threat Timeline Report, which surveys executive perspectives on barriers to action [10].

3. Asset Tokenization: Implementation Advancements

HSBC is the furthest along among major banks in pairing tokenization with quantum defenses. It has trialed quantum-secure technology for distributing tokenized physical gold, using Quantinuum's post-quantum cryptography algorithms and quantum randomness generation, building on its existing Orion tokenization platform and HSBC Gold Token [11]. Separately, HSBC, alongside BT and Toshiba, became the first bank to trial

quantum key distribution (QKD) as a founding member of the London Quantum Secure Metro network [12].

At the more speculative end, sector infrastructure providers are beginning to secure niche tokenized asset classes: Copernic Space and Intrana announced a quantum-resistant blockchain layer that combines post-quantum cryptography with WebAssembly smart contracts, specifically to protect a Real-World Space Asset (RWSA) marketplace covering satellite data licenses and lunar cargo [13].

4. Payment Tokenization: Implementation Advancements

Mastercard has the longest track record here, having approved the first quantum-resistant contactless payment cards under its Enhanced Contactless (Ecos) specification back in October 2022 (manufactured by Giesecke+Devrient and Thales), and has since published a dedicated whitepaper on preparing payment infrastructure for a post-quantum world [14].

A more sobering picture emerges from implementation-tracking analysis: as of early 2026, no hardware security module (HSM) vendor has completed FIPS 140-3 Level 3 CMVP validation that includes PQC algorithms within the validated module boundary, meaning regulated institutions that require this certification cannot yet deploy PQC in production, even though several vendors already ship HSM firmware supporting ML-KEM and ML-DSA [15].

This is a critical bottleneck given that SWIFT mandates FIPS 140-2 Level 2+ (Level 3 for qualified certificates) for its network. On the hardware side, IDEMIA's partnership with GlobalFoundries is targeting mass production in 2026 of a new 28nm smart card chip purpose-built for PQC workloads [15].

On the messaging and settlement side, SWIFT is reportedly piloting on-chain messaging using Linea (an Ethereum Layer-2) and testing a stablecoin-like settlement token, a shift that would embed tokenization directly into interbank messaging infrastructure and raise the stakes for cryptographic migration across that layer [16]. This sits alongside Visa's Tokenized Asset Platform (VTAP) and Mastercard's Multi Token Network (MTN), both of which give banks infrastructure to issue and move tokenized fiat-backed instruments [16].

5. Blockchain / Protocol-Layer Advancements

Below the institutional layer, protocol-level quantum-safe migration tooling is emerging. 01 Quantum's Layer 1 Migration Toolkit (tied to the \$qONE token, launched February 2026) offers a phased model, immediate asset protection via quantum-resistant smart contract wallets, followed by post-quantum architecture design, then full integration with zero-knowledge proofs, explicitly designed to let blockchains mitigate quantum risk without forcing disruptive hard forks [17].

BTQ Technologies reported meaningful 2025 progress in translating quantum-security research

into commercial products: a quantum-safe Bitcoin testnet, live pilots of its QSSN network, and a \$15 million joint investment with ICTK (a South Korean secure-element manufacturer) to co-develop a secure-element platform (QCIM) for hardware-level protection [18].

At the interoperability layer, Chainlink frames its Cross-Chain Interoperability Protocol (CCIP) as a “crypto-agile” connectivity standard designed to absorb future cryptographic upgrades without breaking cross-chain applications, and points to Mind Network’s integration of Fully Homomorphic Encryption (FHE) alongside quantum resistant security for cross-chain transactions as an early proof point [19].

Key Takeaways

Official sector is coordinating, not just warning. BIS Project Leap and the G7 QTWG mark a shift from risk commentary to live technical experimentation and roadmap-setting [1][2][3][4][5].

A large readiness gap persists at the institutional level. The 86%-not-ready statistic from Moody’s/Corinium is the most concrete industry-wide data point available and should be treated as a baseline, not a one-off [8][9].

Certification infrastructure is the current bottleneck, not algorithm availability — the lack of FIPS 140-3 Level 3 PQC validation blocks production deployment for regulated payment operations, even where the technology itself is ready [15].

Individual institutional pilots (HSBC, Mastercard, SWIFT) are ahead of the regulatory/certification curve, suggesting a widening gap between what is technically demonstrated and what is compliant for production use at scale [11][12] [14][15][16].

Protocol-layer tooling is maturing quickly but remains fragmented across vendors (01 Quantum, BTQ, Chainlink/Mind Network), with no dominant standard yet for tokenized-asset chains [17][18][19].

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